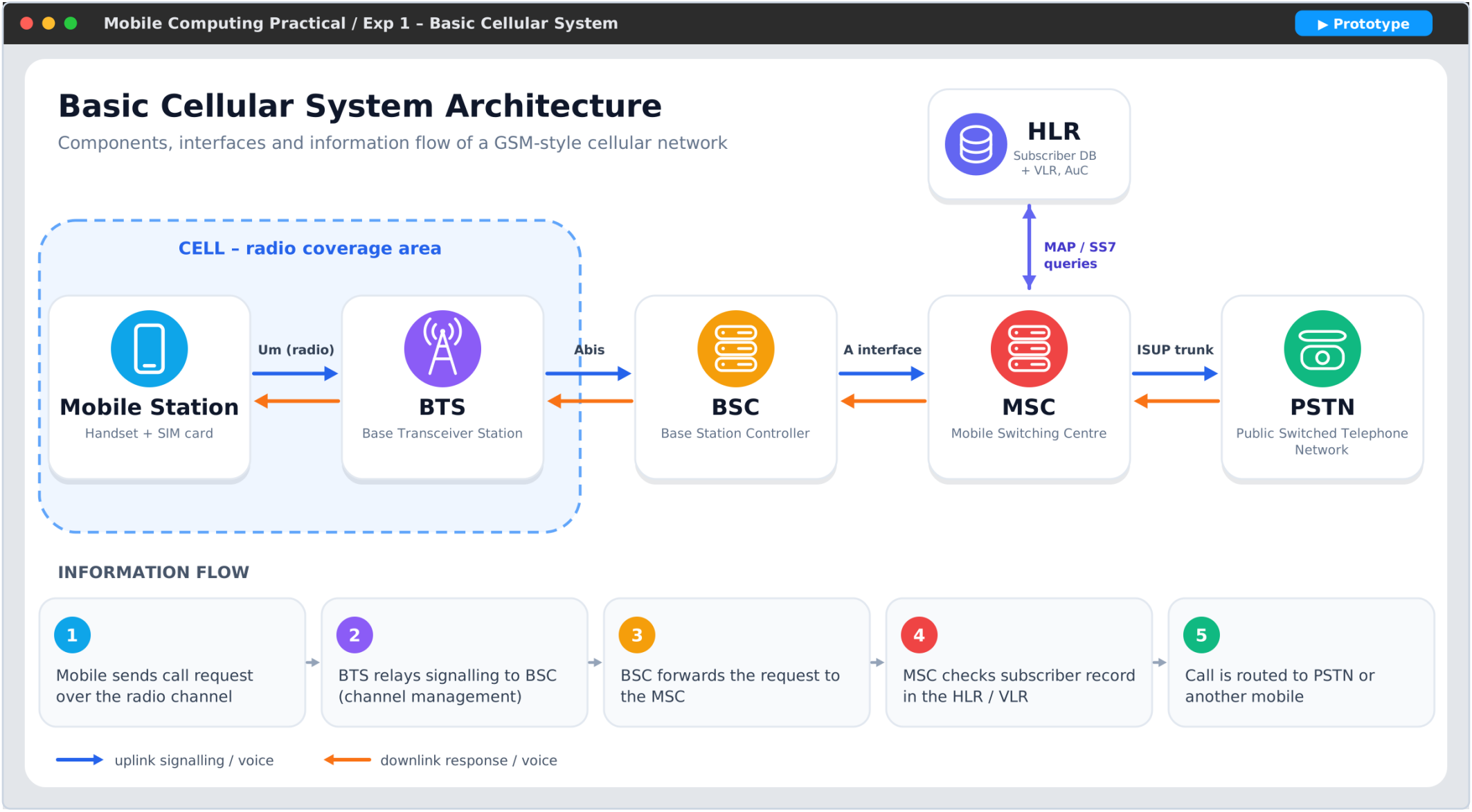
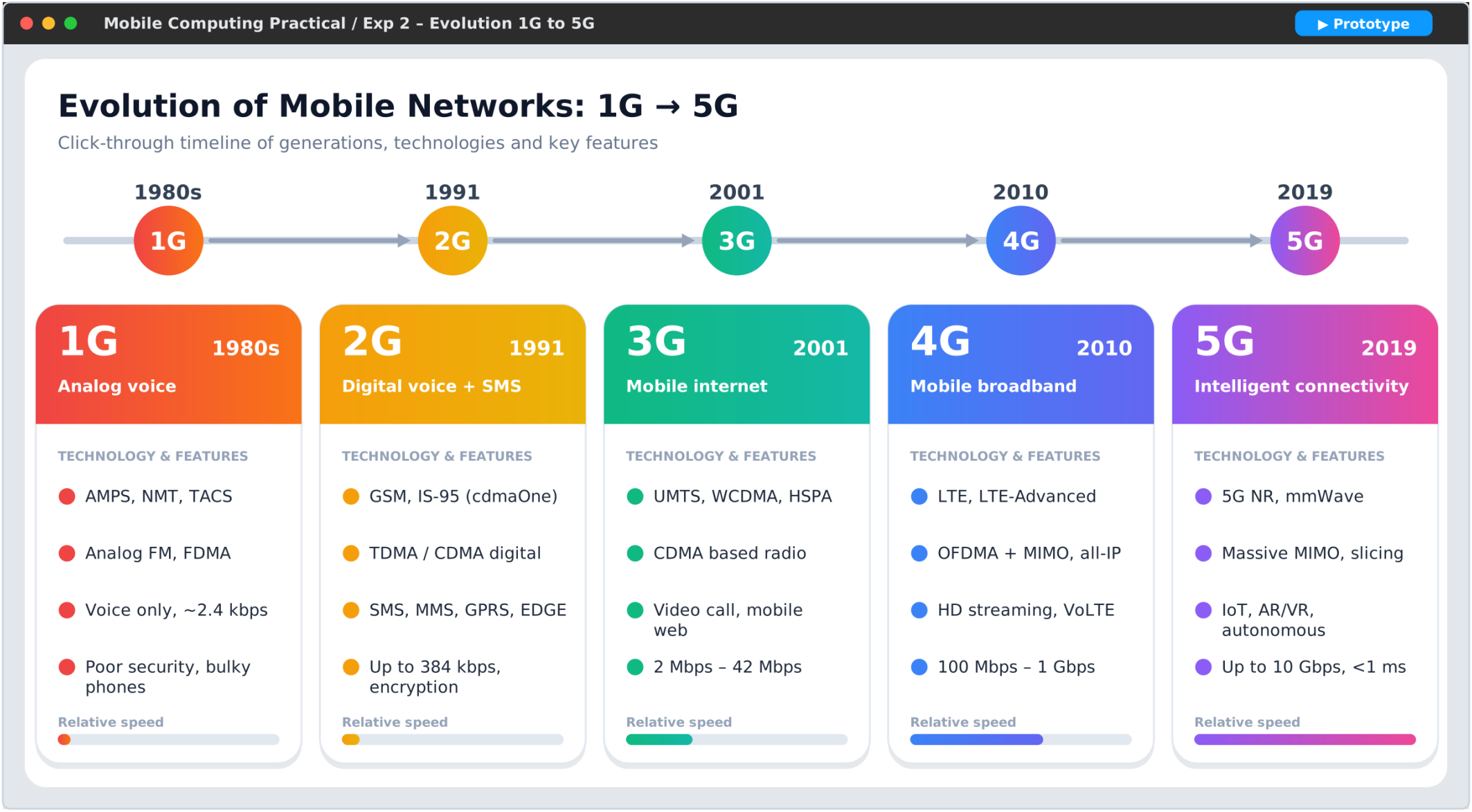
Question 1: Use Figma to create an interactive diagram illustrating the components of a basic cellular system.



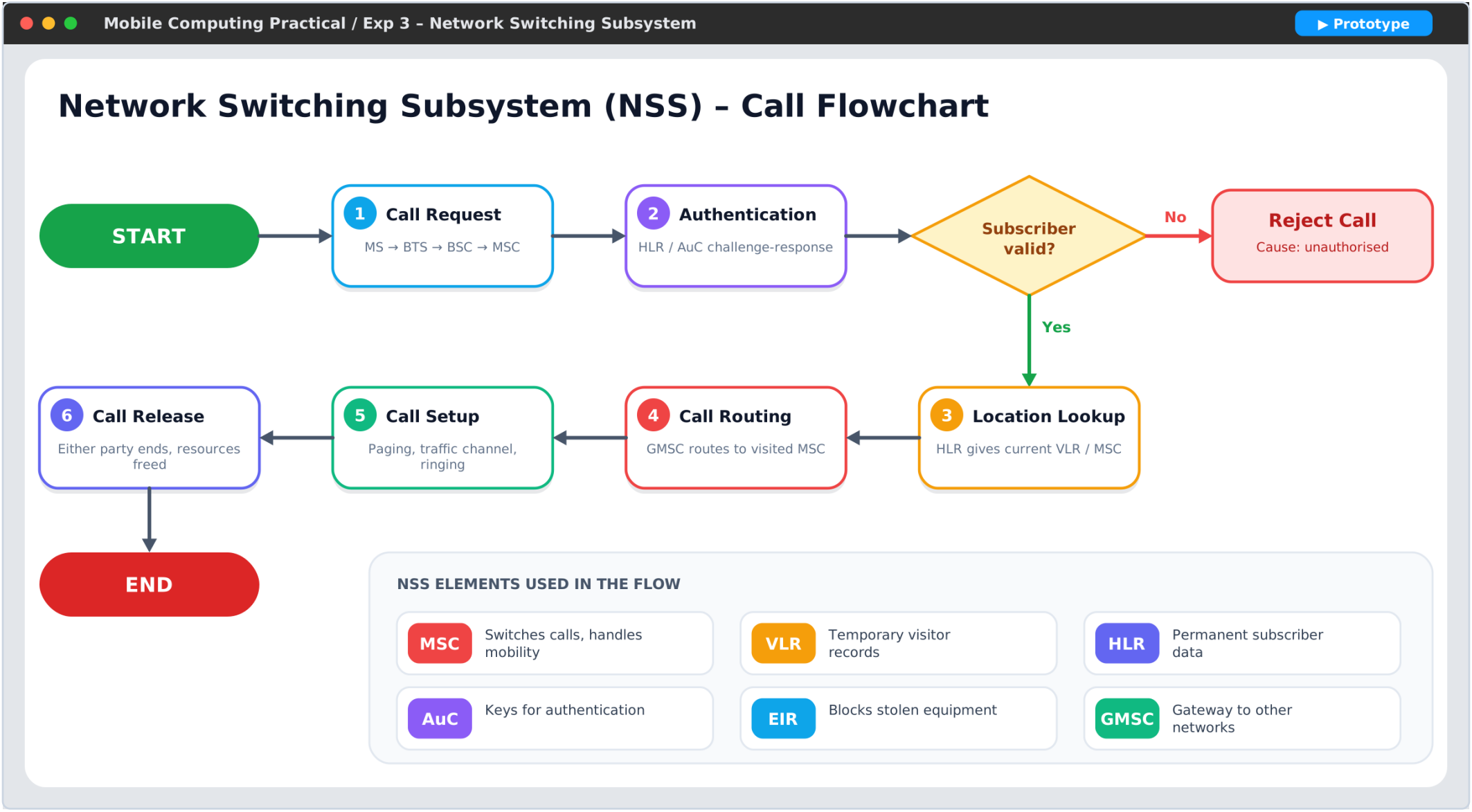
Key labels: Mobile Station, BTS, BSC, MSC, HLR and PSTN with Um, Abis and A interfaces. Blue arrows show uplink signalling, orange arrows the downlink response; the dashed area is one cell.

Question 2: Use Figma to create an interactive diagram illustrating the evolution from 1G to 5G.



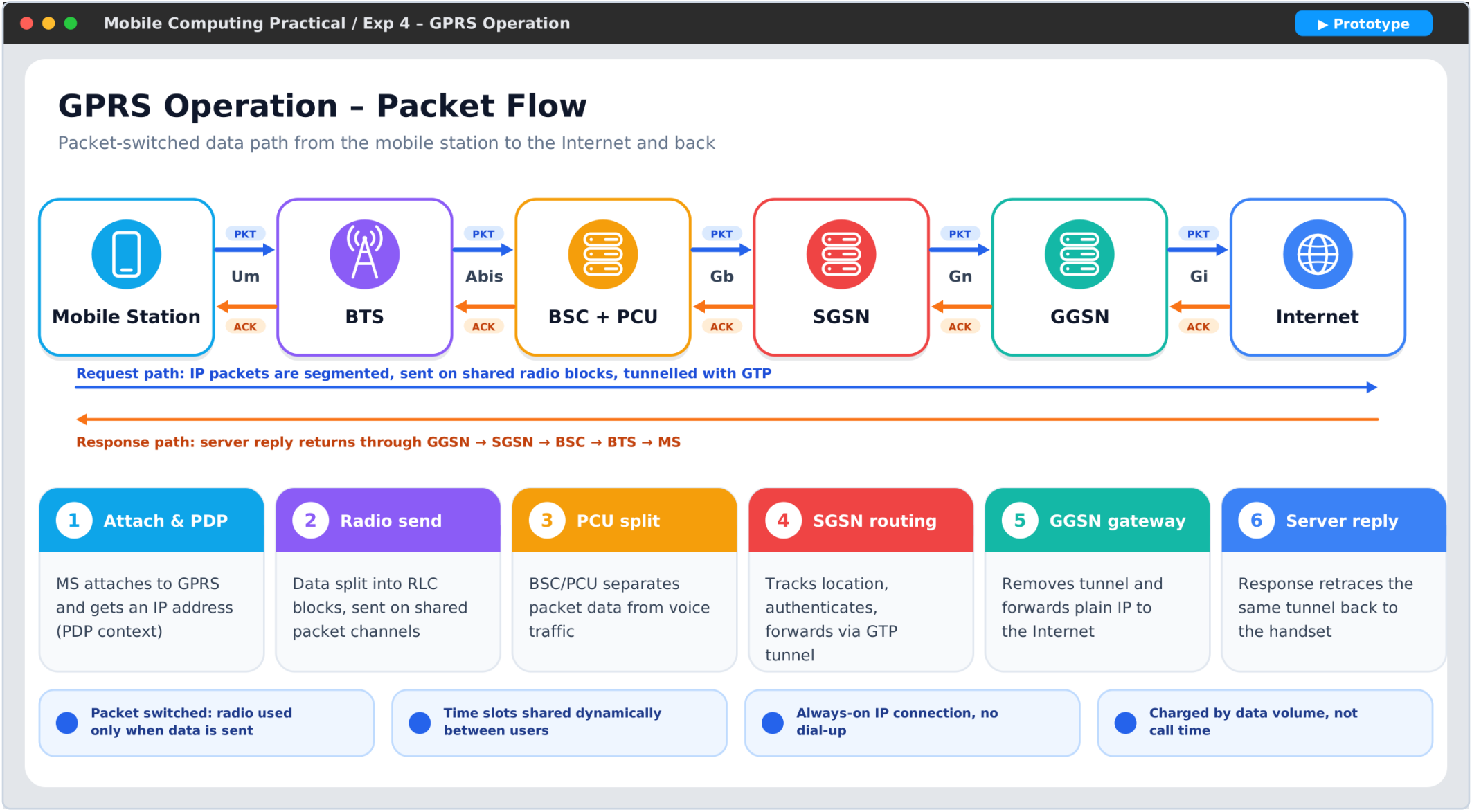
Key labels: Each generation card lists era, key technologies, features and relative speed, moving from analog voice (1G) to intelligent, low-latency connectivity (5G).

Question 3: Develop a flowchart in Figma depicting the operation of a Network Switching Subsystem in a mobile cellular system.



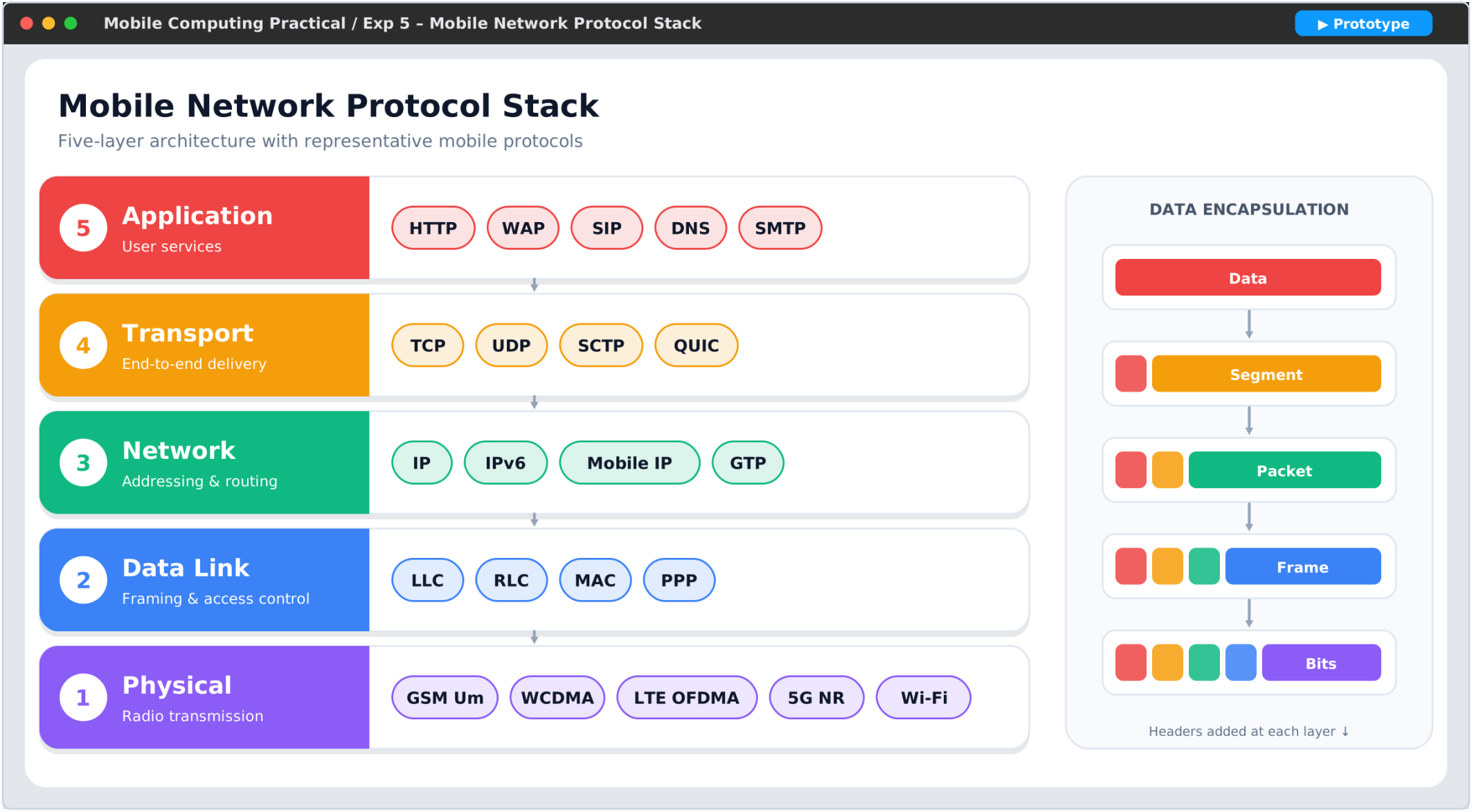
Key labels: Call request → authentication (HLR/AuC) → location lookup → routing via GMSC → call setup → release. A failed authentication branches to call rejection.

Question 4: Develop a flowchart in Figma explaining the operation of General Packet Radio Service (GPRS).



Key labels: Packets flow MS → BTS → BSC/PCU → SGSN → GGSN → Internet (blue) and responses return along the same tunnel (orange). Interfaces: Um, Abis, Gb, Gn, Gi.

Question 5: Create a visual representation in Figma highlighting the key elements of mobile network protocol stacks.



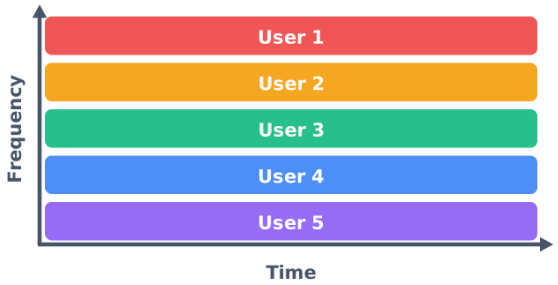
Key labels: Five layers from Application to Physical with example protocols per layer; the side panel shows how each layer adds its header (encapsulation).

Question 6: Create a visual representation in Figma showcasing the differences between various multiple access techniques.

Mobile Computing Practical / Exp 6 – Multiple Access Techniques ▶ Prototype

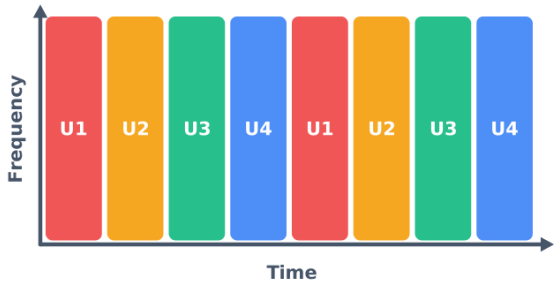
Multiple Access Techniques – Visual Comparison

FDMA — Frequency Division



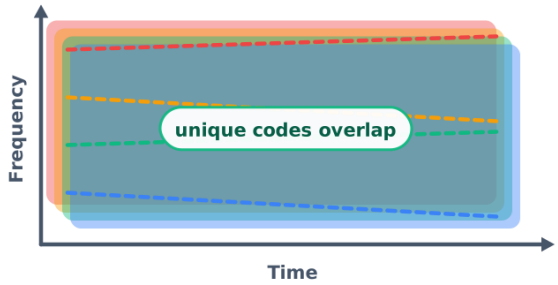
Each user gets its own frequency band for the whole call.

TDMA — Time Division



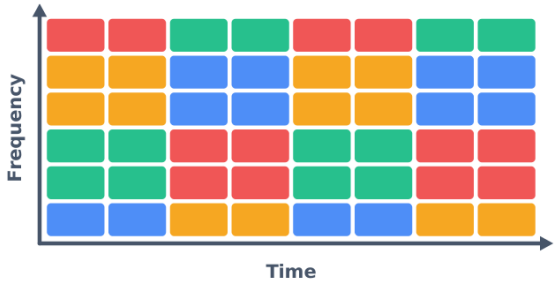
Users share one band, each in a repeating time slot.

CDMA — Code Division



All users transmit together, separated by unique codes.

OFDMA — Orthogonal Frequency Division



Users get sets of sub-carriers and time symbols.

SDMA — Space Division



Directional beams reuse the same resources in space.

AT A GLANCE

	Separates by	Used in
FDMA	Frequency	1G, AMPS
TDMA	Time slot	2G GSM
CDMA	Code	3G WCDMA
OFDMA	Sub-carrier	4G LTE, 5G
SDMA	Space/beam	5G MIMO

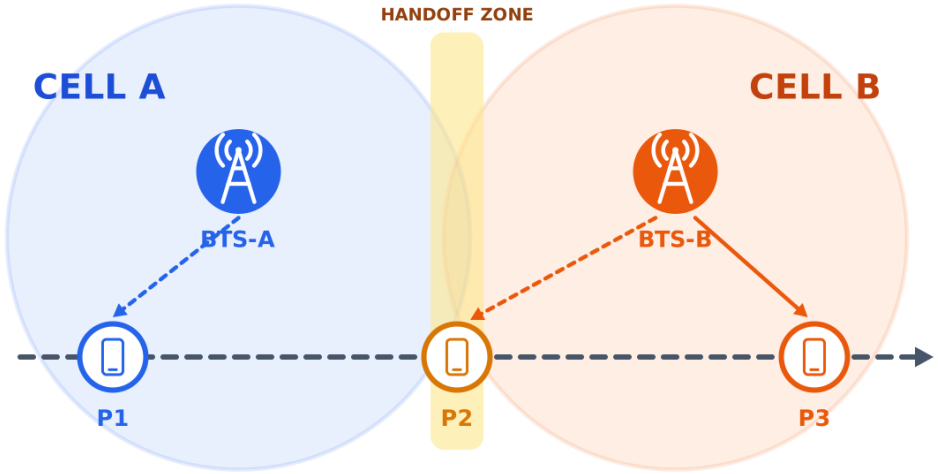
Key labels: FDMA splits by frequency, TDMA by time slot, CDMA by code, OFDMA by sub-carrier and symbol, SDMA by spatial beam. The table summarises generation and usage.

Question 7: Use Figma to simulate a handoff scenario in a mobile network and visualize the process.

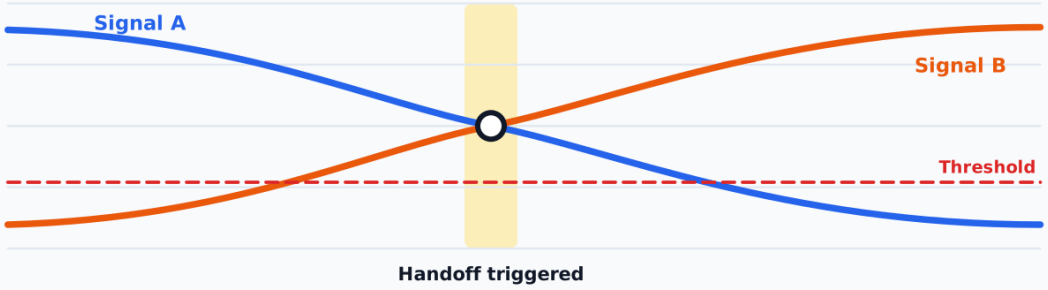
Mobile Computing Practical / Exp 7 – Handoff Simulation ▶ Prototype

Handoff Simulation – Cell A → Cell B

Mobile user moves across the cell boundary while signal strength is monitored



SIGNAL STRENGTH MONITORING

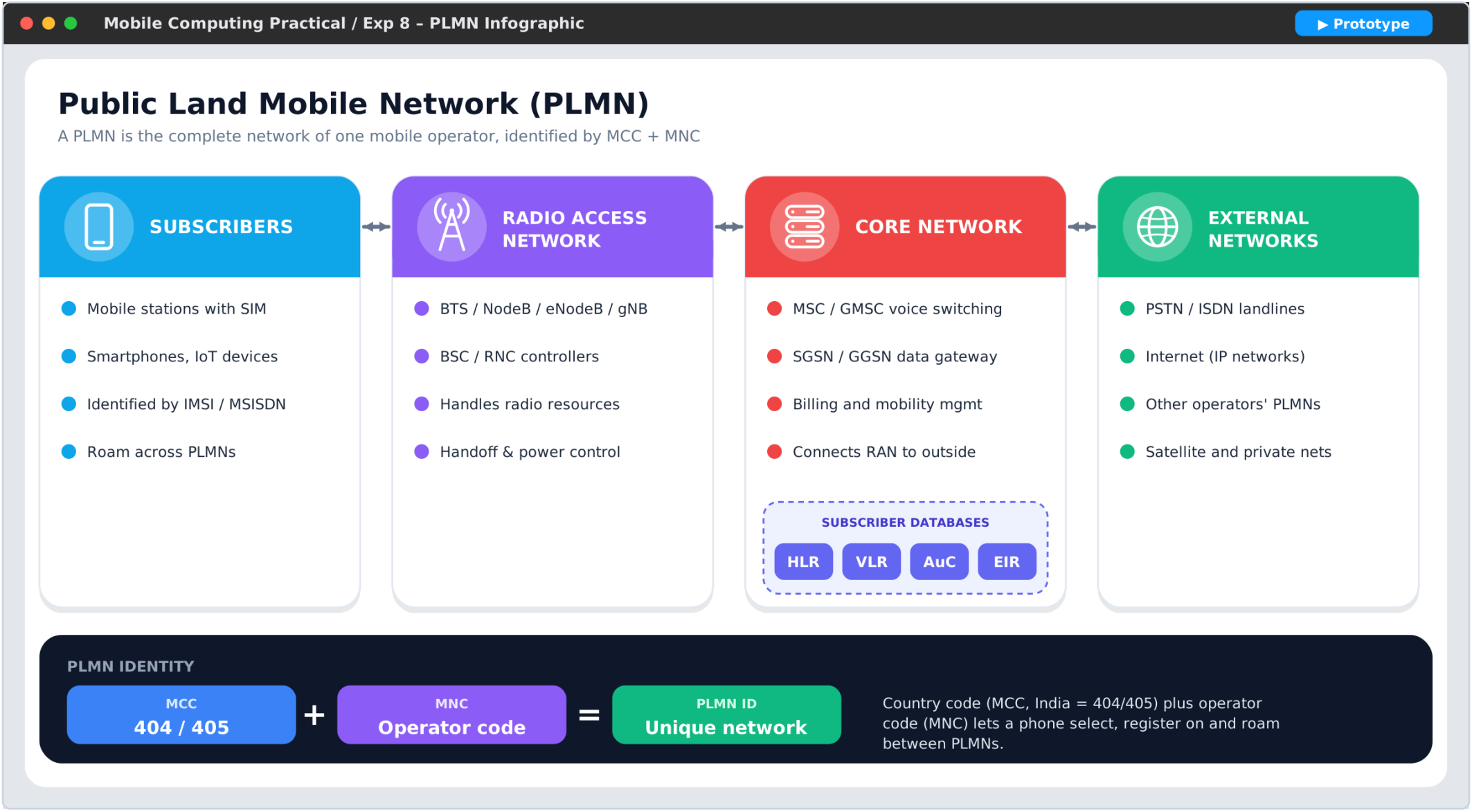


HANDOFF SEQUENCE

1. **Measure** — MS reports signal levels of serving and neighbour cells (MAHO).
2. **Threshold crossed** — Cell A level drops below the threshold while B keeps rising.
3. **Target selection** — BSC/MSC picks Cell B: best level, free channel available.
4. **Handoff command** — Network sends new channel details to the MS via Cell A.
5. **Transfer complete** — MS tunes to Cell B, call continues, old channel released.

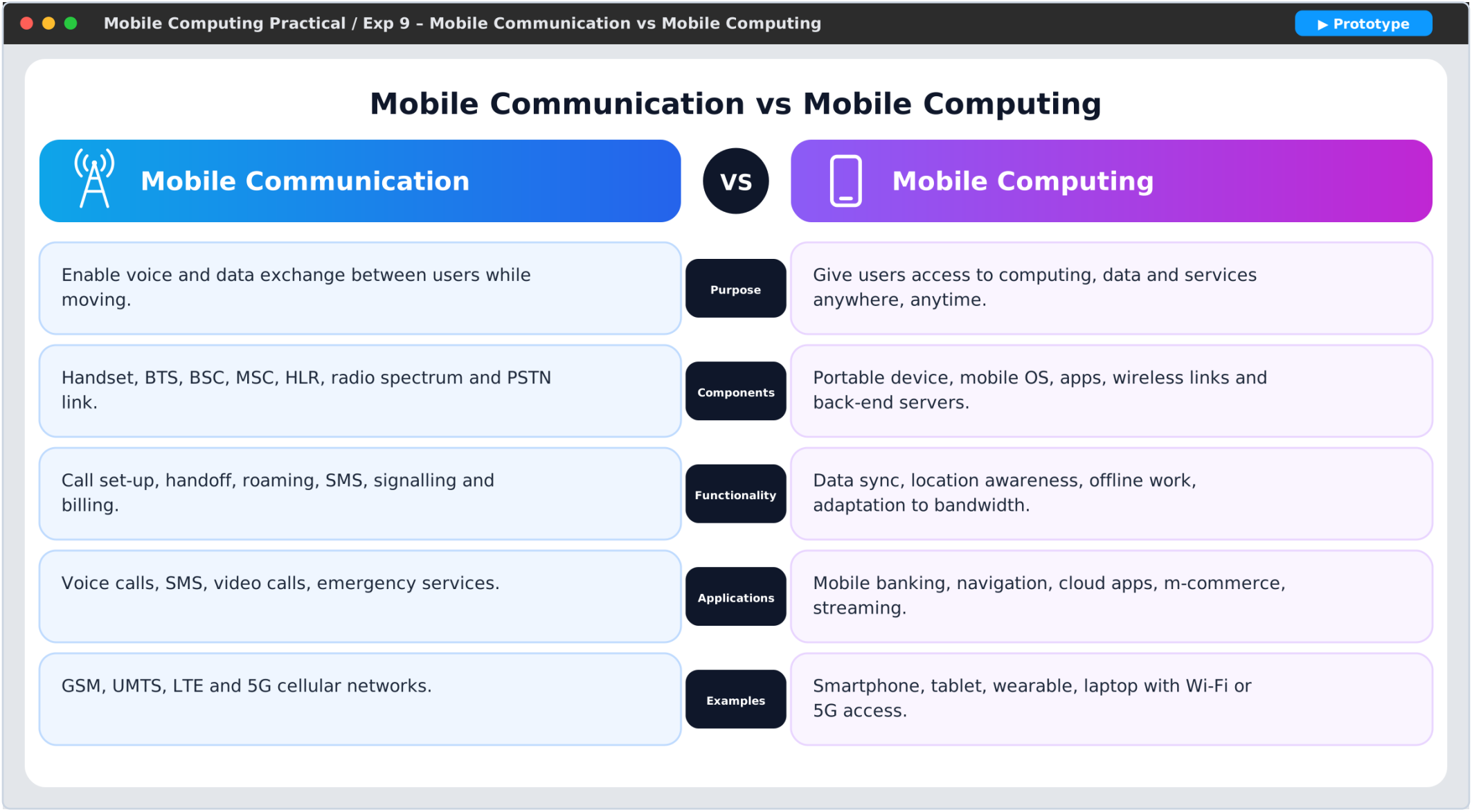
Key labels: A user moves P1 → P2 → P3 from Cell A to Cell B. When signal A falls below the threshold in the handoff zone, the network selects Cell B and transfers the call.

Question 8: Create an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).



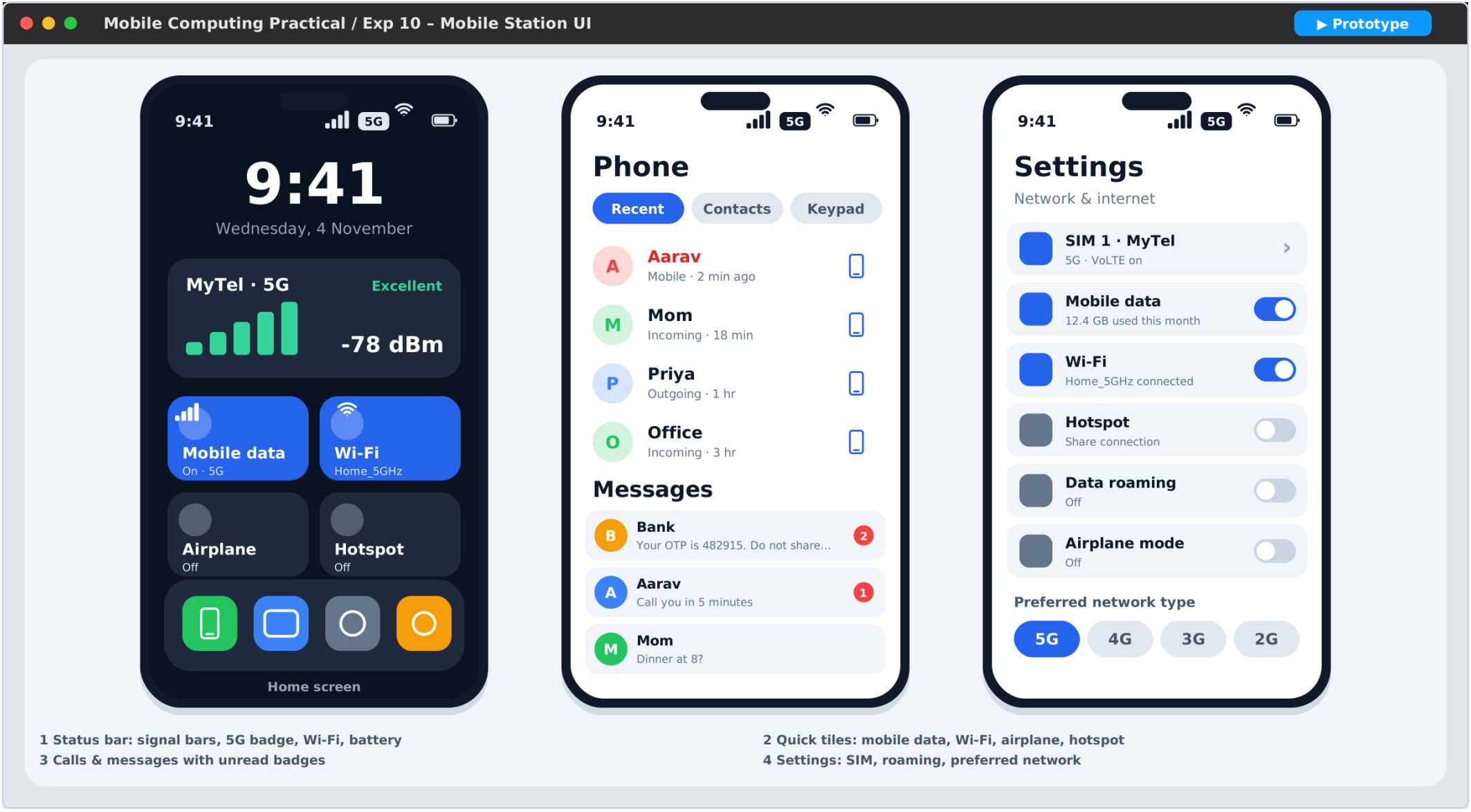
Key labels: Subscribers, radio access network, core network (with HLR, VLR, AuC, EIR) and external networks. PLMN ID = MCC + MNC identifies the operator's network.

Question 9: Design an infographic in Figma comparing mobile communication and mobile computing.



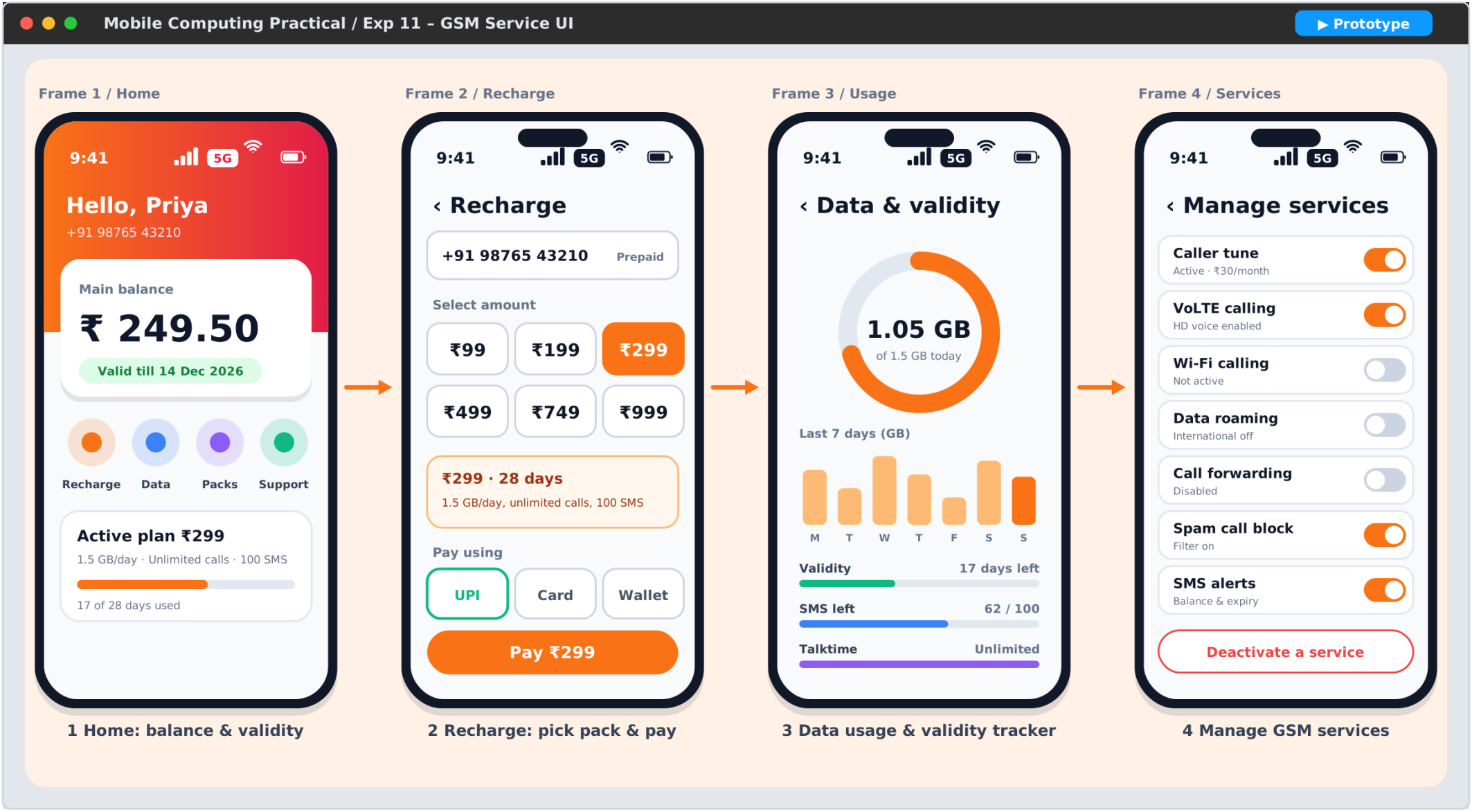
Key labels: Two-column comparison of purpose, components, functionality, applications and examples: the network that carries information vs the computing it enables on the move.

Question 10: Design a user interface prototype for a mobile station using Figma, considering connectivity and convenience.



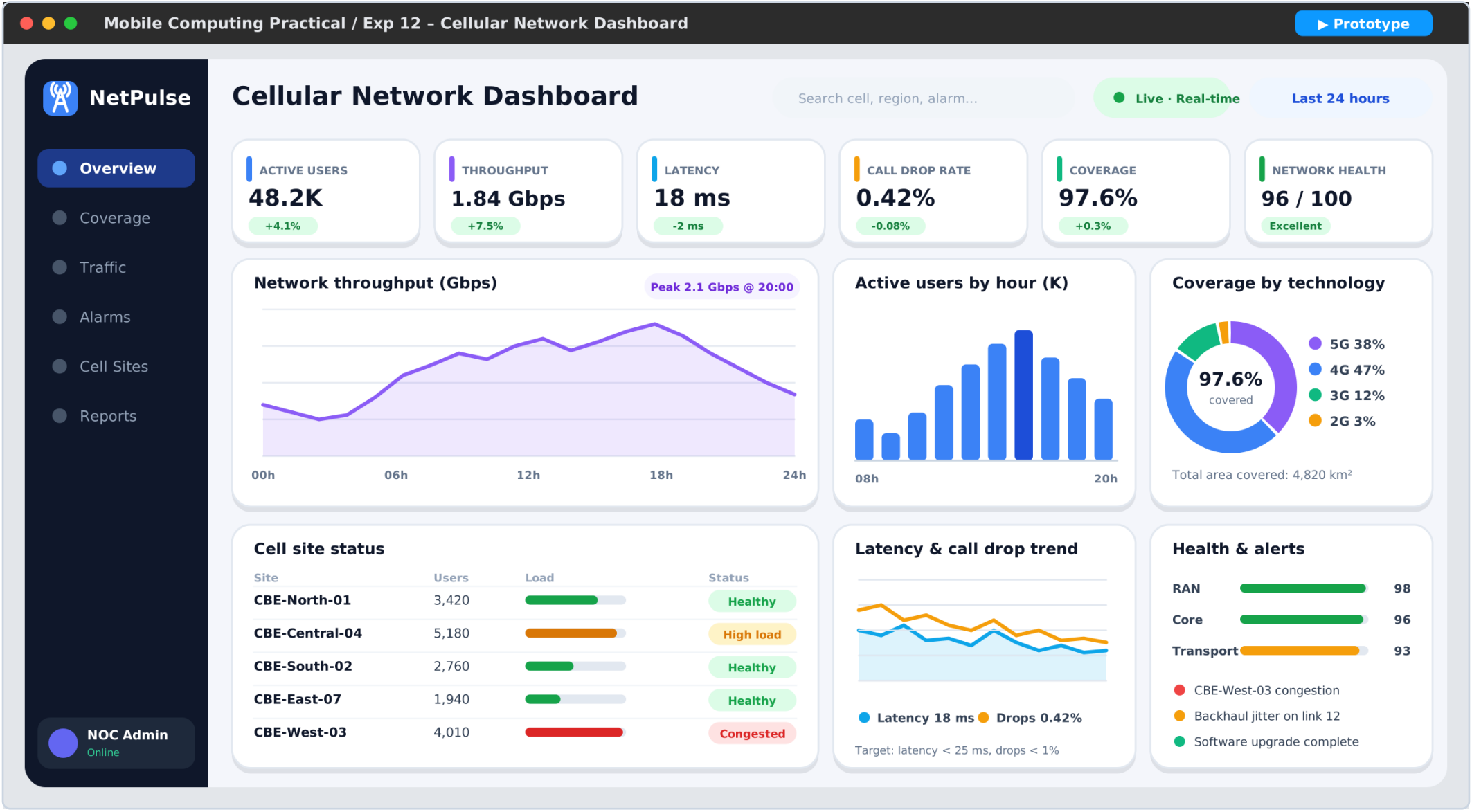
Key labels: Home screen with signal, 5G, Wi-Fi and quick tiles; calls and messages screen with unread badges; settings screen with SIM, roaming and preferred network type.

Question 11: Develop a user interface prototype for a GSM service using Figma.



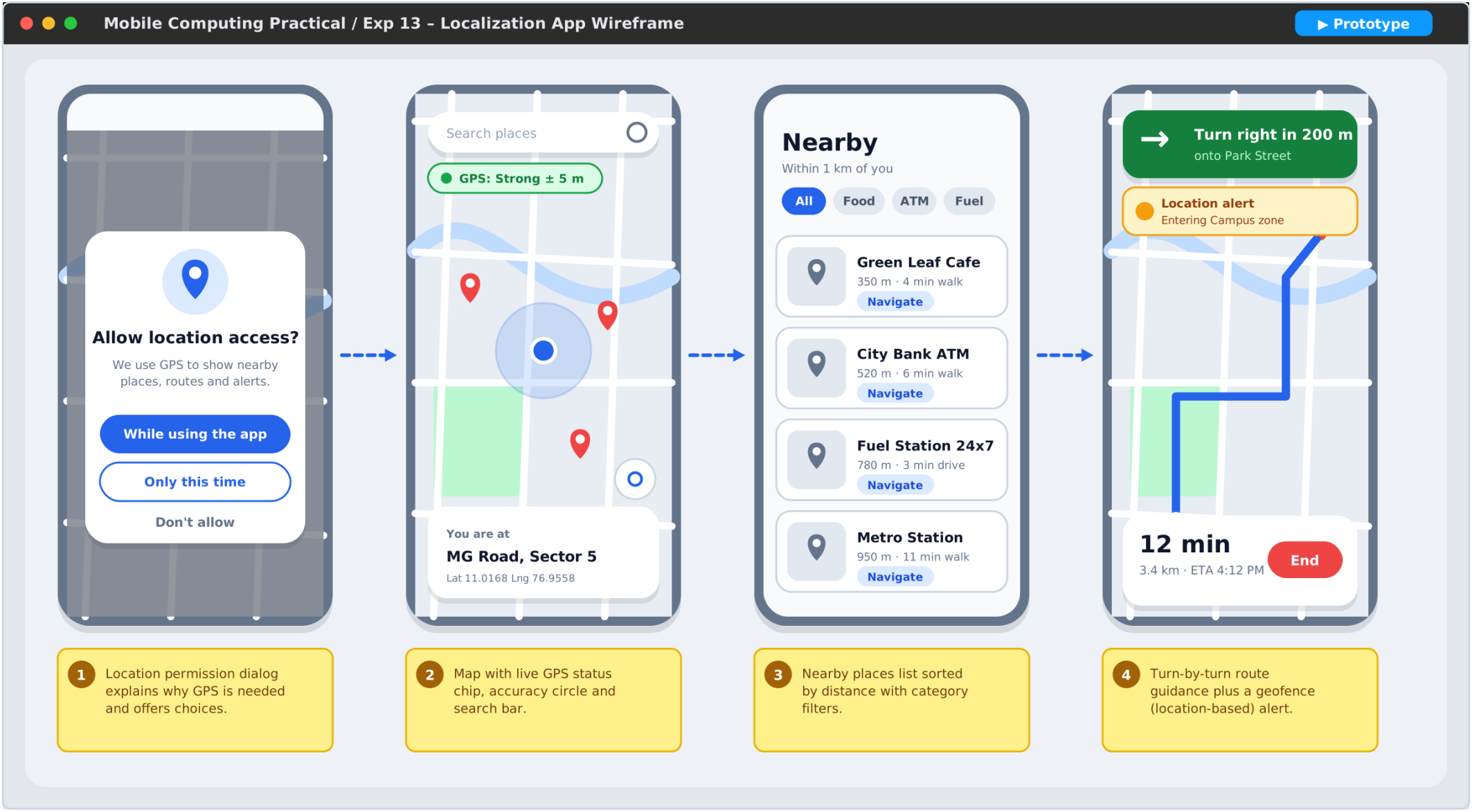
Key labels: Four linked screens: home (balance and validity), recharge, data usage tracker, and service management with toggles. Arrows show the prototype flow.

Question 12: Develop a dashboard prototype in Figma displaying performance criteria for a cellular system.



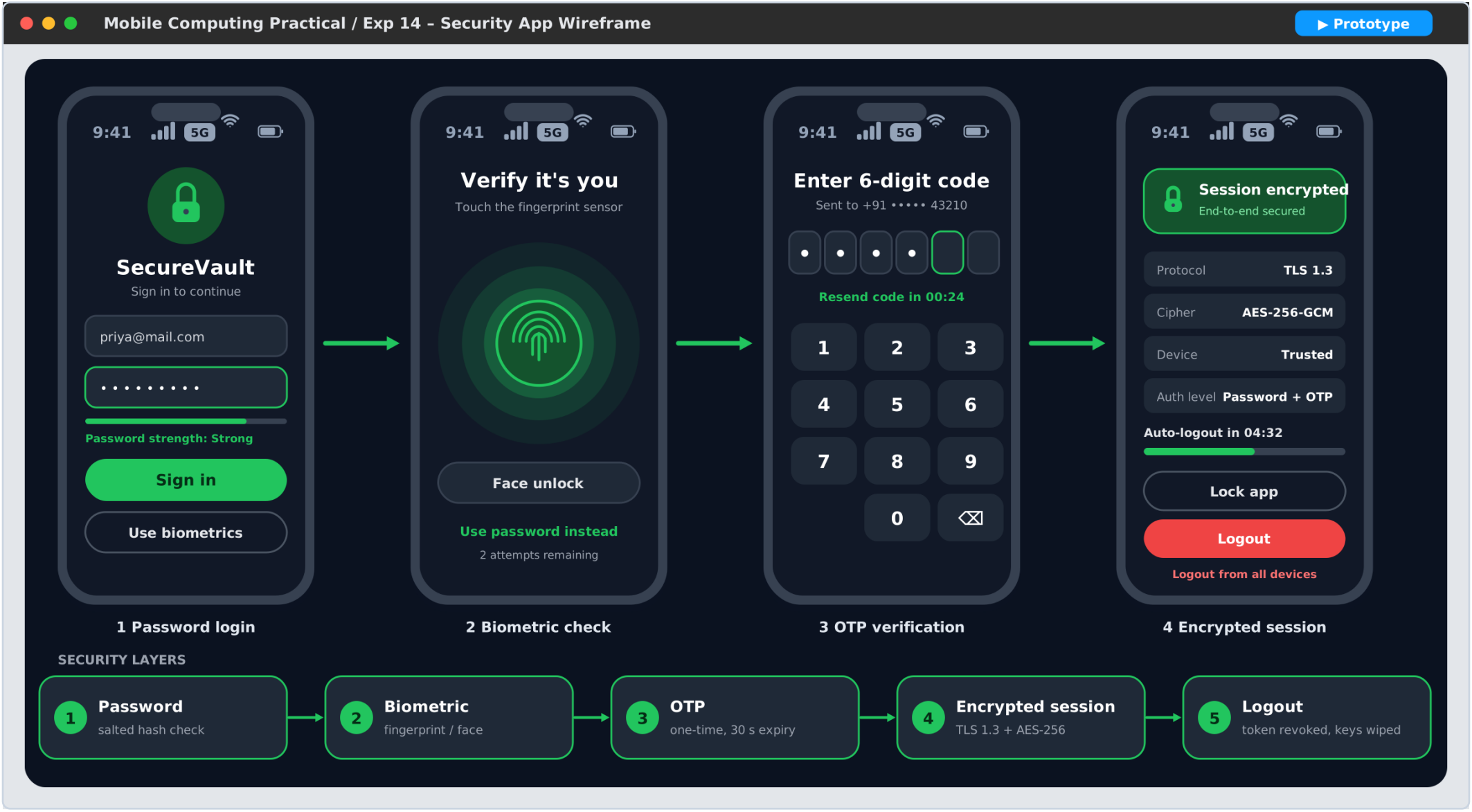
Key labels: KPI cards for users, throughput, latency, call drop rate, coverage and health, plus throughput, user, coverage, latency charts, site table and alerts.

Question 13: Design a wireframe for a mobile application that incorporates localization features using Figma.



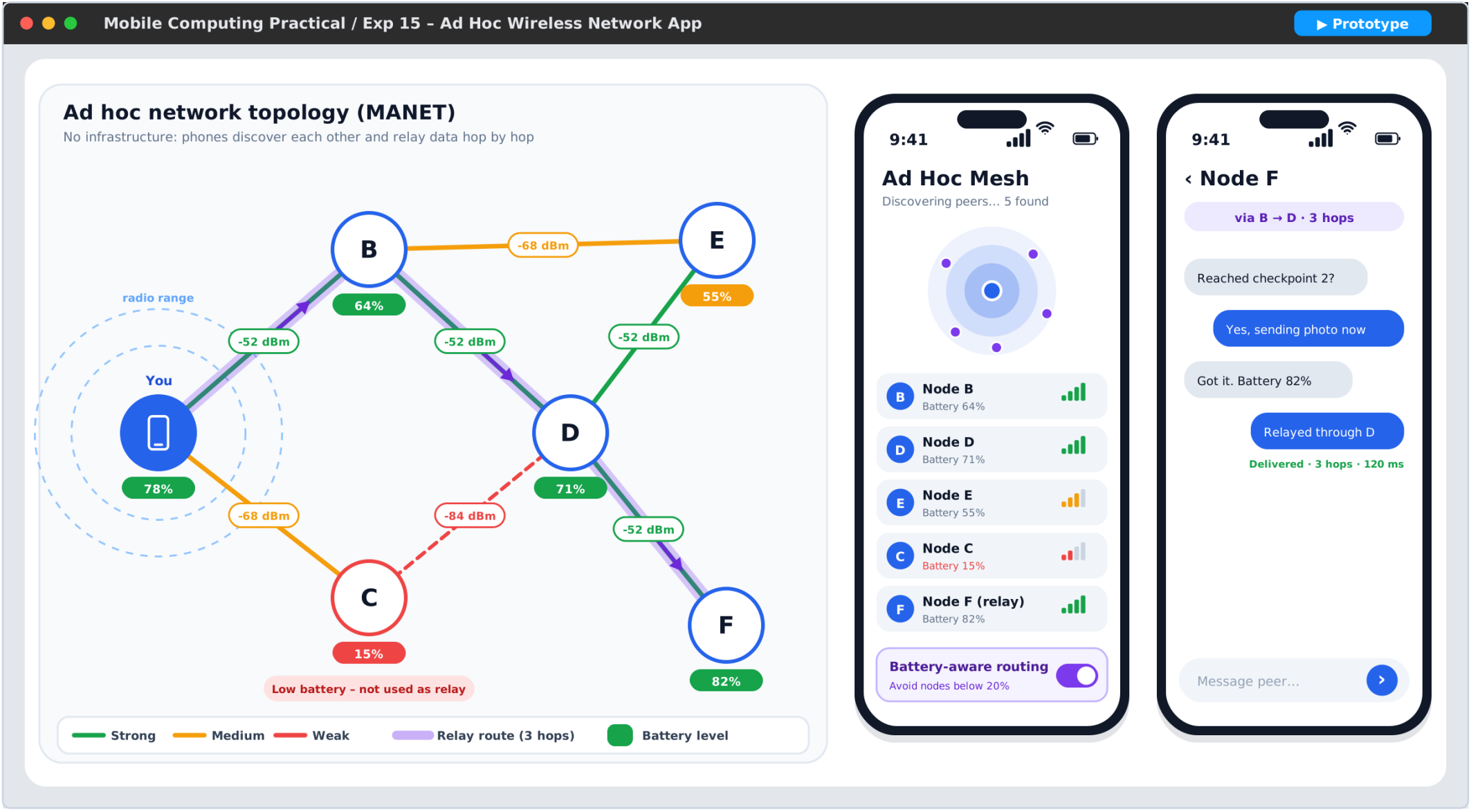
Key labels: Permission dialog, map with GPS status, nearby places list, and route guidance with a geofence alert. Yellow notes explain each screen.

Question 14: Design a wireframe for a mobile application that incorporates security measures using Figma.



Key labels: Password login, biometric check, OTP verification and an encrypted session screen with logout controls; the bottom strip shows the layered security flow.

Question 15: Develop a wireframe for a mobile application that addresses challenges related to ad hoc wireless networks using Figma.



Key labels: Topology with link quality and battery levels, a relay route that avoids a low-battery node, peer discovery with battery-aware mode, and peer messaging over three hops.